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Studierichting: Informatica
Oefeningenreeks 3C nr. 17.14

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 2x + \cos x}{2 + \cos 2x - \sin x} = \frac{0}{0}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{2 \cos 2x - \sin x}{-2 \sin 2x - \cos x} = \frac{-3}{0}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \frac{2 \cos 2x - \sin x}{-2 \sin 2x - \cos x} = \frac{-3}{-0^- - 0^-} = \frac{-3}{0^+} = -\infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \frac{2 \cos 2x - \sin x}{-2 \sin 2x - \cos x} = \frac{-3}{-0^+ - 0^+} = \frac{-3}{0^-} = +\infty$$

References